Odd kernels and cyclotomic modules

The human_compression project

Canonical Topic F: common theory

Abstract

An involution outside an odd normal subgroup is controlled by its orbit under a Klein four complement. In the abelian case, this orbit has two cyclic character components. For nonabelian constructions, quotients of the cyclotomic lattice give explicit finite coordinates, automorphisms and presentation relations. We give the complete elementary arguments and the subgroup intersection tests needed to turn these constructions into regular polyhedra.

1 Marked groups and intersections

We use $x^g = g^{-1}xg$ and $[x, y] = x^{-1}y^{-1}xy$. A rank-three string marking of a finite group G is a generating triple of distinct nonidentity involutions r_0, r_1, r_2 whose endpoints commute. Write $G_I = \langle r_i : i \in I\rangle$ for $I \subseteq \{0, 1, 2\}$. The marking is a *string C-group* if

$$G_I \cap G_J = G_{I\cap J} \qquad (I, J \subseteq \{0, 1, 2\}).$$

Its type is $\{k_1, k_2\}$, where $k_1 = |r_0r_1|$ and $k_2 = |r_1r_2|$. We call it nondegenerate when $k_1, k_2 \geq 3$, and tight when $|G| = 2k_1k_2$. A marked isomorphism preserves the three generators; duality reverses them. We retain the classical correspondence between string C-groups and abstract regular polyhedra when using geometric terminology.

Lemma 1 (Rank-three intersection test)**.** *A nondegenerate rank-three string marking is a string C-group if and only if*

$$\langle r_0, r_1\rangle \cap \langle r_1, r_2\rangle = \langle r_1\rangle. \tag{1}$$

In that case $2k_1k_2 \leq |G|$.

Proof. Necessity is part of the intersection property. For sufficiency set $H_{ij} = \langle r_i, r_j\rangle$. The endpoint group consists of $1, r_0, r_2, r_0r_2$. If its intersection with H_{01} contains r_2 or r_0r_2, then $r_2 \in H_{01}\cap H_{12} = \langle r_1\rangle$, contrary to distinctness. Hence $H_{02} \cap H_{01} = \langle r_0\rangle$; dually $H_{02} \cap H_{12} = \langle r_2\rangle$. These equalities and (1) also exclude a generator from the group on the other two. Singleton intersections follow, and pairs involving the empty or full index set are immediate. This accounts for every pair of subsets. Finally $|H_{01}| = 2k_1$ and $|H_{12}| = 2k_2$, so

$$|H_{01}H_{12}| = \frac{|H_{01}|\,|H_{12}|}{|H_{01} \cap H_{12}|} = 2k_1k_2 \leq |G|.$$

□

Lemma 2 (Prime-index intersection)**.** *Let C, H, J be subgroups of a group, with $C \leq H \cap J$ and $[H : C] = p$ prime. If $H \not\leq J$, then $H \cap J = C$. In particular, this holds if $\langle H, J\rangle = G$ and $J < G$.*

Proof. The index tower gives $p = [H : H \cap J][H \cap J : C]$. The first factor is not one, so the second is one. In the final situation $H \leq J$ would give $G = \langle H, J\rangle = J$, contrary to the hypothesis. □

For later products it is useful to record the elementary identities

$$(sh)^2 = 1, \qquad (sh)(bh) = sb^{-1}, \tag{2}$$

whenever $h^2 = 1$, $hsh = s^{-1}$ and $hbh = b^{-1}$. The first equality uses only the condition involving s; the second uses only the condition involving b.

2 An inverted orbit and its characters

Lemma 3 (One inverted orbit)**.** *Suppose $G = P \rtimes K$, where P has odd order and $K = \langle a, b\rangle$ is a Klein four group. Suppose $G = \langle K, r\rangle$, where r is a nonidentity involution. Write $r = ut$ with $u \in P$, $t \in K$. Then $t \neq 1$, $u^t = u^{-1}$, and*

$$P = \langle u, u^k\rangle \qquad \text{for every } k \in K \setminus \langle t\rangle.$$

Proof. An odd-order group has no nonidentity involution, so $t \neq 1$. The equation $(ut)^2 = 1$ gives $u^t = u^{-1}$. Put $L = \langle u, u^k\rangle$. Since t commutes with k, it inverts both displayed generators of L. The element k exchanges them. Thus L is K-invariant and LK is a subgroup containing K and r. It equals G, and intersecting with P gives $L = P$. □

This proof says that t inverts the two indicated generators. It does not assert that inversion is an automorphism of a nonabelian group.

Lemma 4 (Two cyclic characters)**.** *In Lemma 3, suppose P is abelian and noncyclic. Then it is the direct product of two nontrivial cyclic subgroups P_χ, indexed by the two characters $\chi : K \longrightarrow \{1, -1\}$ with $\chi(t) = -1$. The projection of u generates each P_χ.*

Proof. Write P additively. Multiplication by four is an automorphism of P. The four endomorphisms

$$e_{\epsilon,\delta} = \tfrac{1}{4}(1 + \epsilon a)(1 + \delta b), \qquad \epsilon, \delta \in \{1, -1\},$$

are orthogonal idempotents with sum the identity: expand the products using $a^2 = b^2 = 1$ and $ab = ba$. Their images give the four character components, on which a, b act by ϵ, δ respectively. The element t inverts u and commutes with K, so it inverts the whole K-orbit of u. That orbit generates P by Lemma 3; hence t acts as -1 on P. Every component with $\chi(t) = 1$ is therefore annihilated by two and is zero. For either remaining character, the projection of each orbit element is plus or minus $e_\chi u$. Its component is consequently cyclic. If one component were zero then P would be cyclic, so both are nontrivial. □

3 The complete cyclotomic lattice

Let p be an odd prime, let ζ be a primitive pth root of unity, and put $O = \mathbb{Z}[\zeta]$ and $\pi = \zeta - 1$.

Theorem 5 (Cyclotomic coordinates)**.** *For every $n \geq 1$, the ring $A_n = O/\pi^n O$ has p^n elements. Write uniquely*

$$n = (e-1)(p-1) + r, \qquad e \geq 1, \quad 1 \leq r \leq p-1.$$

As an additive group with the indicated ordered generators,

$$A_n = \langle 1\rangle \oplus \langle \pi\rangle \oplus \cdots \oplus \langle \pi^{p-2}\rangle \cong (\mathbb{Z}/p^e\mathbb{Z})^r \oplus (\mathbb{Z}/p^{e-1}\mathbb{Z})^{p-1-r}. \tag{3}$$

A modulus-one factor is trivial. Complex conjugation induces an involutory automorphism of A_n. Multiplication by ζ has order p when $n \geq 2$, and for $j \not\equiv 0 \pmod p$ it satisfies $\sum_{i=0}^{p-1} \zeta^{ij} = 0$.

Proof. The polynomial

$$\Phi_p(1+X) = p + \binom{p}{2}X + \cdots + \binom{p}{p-1}X^{p-2} + X^{p-1}$$

is Eisenstein at p. Thus O is an integral domain with $\mathbb{Z}$-basis $1, \pi, \ldots, \pi^{p-2}$. Substitution $\zeta = 1$ gives $O/\pi O \cong \mathbb{F}_p$. Multiplication by π^j induces an isomorphism $O/\pi O \longrightarrow \pi^j O/\pi^{j+1}O$: surjectivity is immediate and injectivity follows by cancellation in the domain. The successive indices are all p, whence

$$|O/\pi^n O| = p^n. \tag{4}$$

The identity $p = \prod_{j=1}^{p-1}(1-\zeta^j)$ gives $pO = \pi^{p-1}O$. Here equality, rather than inclusion alone, follows because each ratio $(1-\zeta^j)/(1-\zeta)$ is a unit of O. It belongs to O by a geometric sum. The automorphism $\zeta \mapsto \zeta^j$ identifies the norms of numerator and denominator, both equal to p, so the ratio has norm one. Its multiplication matrix on the integral basis has determinant one; the adjugate gives an integral inverse and hence a unit of O.

For $1 \le r \le p-1$ define the integral lattice

$$L_r = p\mathbb{Z}\,1 + \cdots + p\mathbb{Z}\,\pi^{r-1} + \mathbb{Z}\,\pi^r + \cdots + \mathbb{Z}\,\pi^{p-2}.$$

Every displayed generator is in $\pi^r O$, using $pO = \pi^{p-1}O$ for the first group of generators. Both L_r and $\pi^r O$ have index p^r in O, the latter by (4); thus $L_r = \pi^r O$. Now $\pi^n O = p^{e-1}\pi^r O$, which gives every coordinate modulus in (3). In particular it gives the individual orders, including trivial listed generators when $e = 1$.

The equation $\overline{\pi} = -\zeta^{-1}\pi$ shows that conjugation preserves $\pi^n O$. Multiplication by ζ has pth power the identity. For $n \ge 2$ it is not the identity, since its difference from the identity sends 1 to $\pi \notin \pi^n O$; the latter follows from the strict successive indices above. Its order is therefore p. Finally $1 + \zeta + \cdots + \zeta^{p-1} = 0$; multiplication of exponents by a nonzero residue j permutes these terms. □

Corollary 6 (Conjugation and displacement)**.** *Let A_n be written additively and let an element β of order p act by $v^\beta = \zeta v$. Put $s_i = \pi^{i-1}$ for $i \ge 1$. Then $[s_i, \beta] = s_{i+1}$ in $A_n \rtimes \langle\beta\rangle$, and*

$$\overline{s_i} = (-1)^{i-1}\zeta^{1-i}s_i, \qquad i \ge 1, \tag{5}$$

$$\zeta^{1-i}s_i = \sum_{j=i}^{p+1}\binom{p+1-i}{j-i}s_j, \qquad 1 \le i \le p. \tag{6}$$

The first expression on the left of the commutator identity is group notation; the two displayed equations are additive ring identities.

Proof. The commutator is $(\zeta - 1)s_i = \pi s_i$. Equation (5) follows by raising $\overline{\pi} = -\zeta^{-1}\pi$ to the power $i-1$. For (6), use $\zeta^{1-i} = \zeta^{p+1-i}$ and expand $(1+\pi)^{p+1-i}\pi^{i-1}$. No independence of the additional s_p, s_{p+1} is asserted or needed. □

4 Attribution and formal scope

The rank-three criterion is standard and appears as Proposition 2.1 of Hou, Feng, Leemans and Qu [1]. The inverted-orbit argument reorganizes their structural proof; the character projectors make explicit the classical decomposition used in their Lemma 2.4. The cyclotomic arithmetic is classical. Its use here supplies one proof of the order, coordinates and automorphisms of the presentation used in their Section 5. We make no historical priority claim for that organization.

The accompanying application supplies all source-specific proofs. In its tightness converse it retains the stated odd/even classification of Cunningham and Pellicer; that input is not a conclusion of

this common paper. Sylow's theorem, Eisenstein's criterion, elementary finite group theory and the usual determinant norm are standard background. Every ideal and coordinate calculation used here is proved here.

The unchanged Lean module `OddKernelIntersection` verifies five statements: the prime-index intersection lemma and its generation variant, two reflection identities, and the square of the literal incorrect marking discussed in the application. It does not formalize the orbit, character or cyclotomic arguments, or the complete rank-three criterion. The written proofs of those statements remain part of this paper's mathematical dependency boundary.

References

[1] D.-D. Hou, Y.-Q. Feng, D. Leemans and H.-P. Qu, *String C-groups of order* $4p^m$, arXiv:2407.10388v1, 15 July 2024, 12 pp.